Accepted Date: 05-Aug-2014

Article Type: Original Article

Email id : dmckenzie@worldbank.org

The Impact of Vocational Training for the Unemployed: Experimental Evidence from Turkey[#]

SHORT TITLE: **Impact of Vocational Training in Turkey**

Sarojini Hirshleifer

David McKenzie

Rita Almeida

Cristobal Ridao-Cano

Abstract

We use a randomized experiment to evaluate Turkey's vocational training programs for the unemployed. A detailed follow-up survey of a large sample with low attrition enables precise estimation of treatment impacts and their heterogeneity. The average impact of training on employment is positive, but close to zero and statistically insignificant, which is much lower than program officials and applicants expected. Over the first year, training had statistically

[#] We thank the Spanish Impact Evaluation Fund (SIEF), the Gender Action Plan, the World Bank's Research Support Budget, and ISKUR for co-funding this impact evaluation. We thank Ayca Donmez, Elcin Koc, and Levent Yener for their assistance on this project, the editor, two anonymous referees, and audiences at a number of seminars for useful comments, and ISKUR for their collaboration. All views expressed in this paper are those of the authors alone and need not necessarily reflect those of the World Bank or ISKUR. Corresponding author: David McKenzie, The World Bank, MSN MC3-307, 1818 H Street NW, Washington DC 20433, USA.

This article has been accepted for publication and undergone full peer review but has not been through the copyediting, typesetting, pagination and proofreading process, which may lead to differences between this version and the Version of Record. Please cite this article as doi: 10.1111/ecoj.12211

significant effects on the quality of employment, and these positive impacts are stronger when training is offered by private providers. However, administrative data shows that after three years these effects have also dissipated.

Keywords: Vocational training; Active Labour Market Programs; Randomized Experiment; Private Provision.

JEL Codes: I28, J24, J68, O12, C93.

After a decline in funding in the 1990s and early 2000s, vocational training programs began to return more prominently to the agendas of governments and international donor agencies in the mid-2000s (King and Palmer, 2010). An employer-identified unmet demand combined with constraints on the supply of skills in the working-age population has created a concern that low skill levels are impeding development in some countries (UNESCO, 2012; World Bank 2012a). The global economic recession that began in 2007 has also dramatically increased interest in policies that could be used to reduce unemployment (World Bank, 2012b). The persistence of labour market imbalances has led to the worry that unemployment is becoming more structural in nature, requiring an emphasis on skills training to help reduce skills mismatches (ILO, 2012). As a result, expanded training programs were the most common type of labour market policy implemented globally in response to the crisis (McKenzie and Robalino, 2010).

The key question is then: do such policies work in helping individuals who receive training to subsequently find jobs? A review of the U.S. literature by Heckman et al. (1999) found substantial heterogeneity in the estimates of impact across studies and concluded that job training had at most modest positive impacts on adult earnings, with no impact or even negative impacts for youth.[1] Similarly, Kluve (2010) in reviewing evaluations of programs in Europe concludes

[1] More recently, Schochet et al. (2008) find improvements in earnings for disadvantaged youth taking part in the Jobs Corps program.

that they also show, at best, modest positive effects, but that programs for youth are less likely to show positive impacts. But most of these programs have been offered to especially disadvantaged groups or targeted only at youth: we are not aware of an experimental evaluation of an at-scale vocational training program for the unemployed in developed countries.

In developing countries, there have been few rigorous evaluations of training programs. Job training may be more effective in developing countries, however, if a skills gap is especially likely to be the binding constraint to employment (Dar et al, 2004; World Bank 2012a). Recently, three randomized evaluations have been conducted of vocational training programs directed at disadvantaged youth in Colombia (Attanasio et al, 2011), the Dominican Republic (Card et al, 2011), and Malawi (Cho et al, 2013). The results in Malawi and the Dominican Republic are consistent with the earlier literature, with no impact on employment in either, and perhaps modest increases in income in the Dominican Republic. Somewhat more encouraging results are found in Colombia with young women offered training having a 7% increase in employment and 20% increase in earnings, although men saw no change in these outcomes.[2]

This paper builds on and extends this literature by providing the first randomized experiment of a large-scale vocational training program for the general unemployed population (not just for disadvantaged youth) in a developing country. This is also the first paper on a developing country that is able to trace longer-term impacts up to three years post-training, by complementing a follow-up survey with administrative data from the social security agency.. We employ an over-subscription design to evaluate the impact of the Turkish National

[2] In addition a small pilot study of 658 women offered tailoring and stitching training in India found a 5 percent increase in employment (Maitra and Mani, 2012), while two studies of youth in Uganda which offer vocational training in combination with a grant (Blattman et al, 2014) or in combination with life skills training (Bandiera et al, 2012) also found positive impacts on employment and/or earnings.

Employment Agency's vocational training programs.[3] These programs average 336 hours over three months, are available for a wide range of subjects, and are offered by both private and public providers. These training services were provided to over 250,000 registered unemployed in 2011, hence we are evaluating a program operating at scale and not just a pilot. A large sample of 5,902 applicants randomly allocated to treatment and control within 130 separate courses, coupled with a detailed follow-up survey with only 6% attrition one year after training allows us to measure both the overall impact of training, and heterogeneity in training impacts along dimensions pre-specified in a pre-analysis plan. These features contrast with the small existing literature in developing countries, which has focused on pilot programs for youth, evaluated over relatively short time horizons, with higher rates of attrition, and no sources of administrative data on employment outcomes.

We estimate that being assigned to training had an overall effect on employment and earnings that was small in magnitude and not statistically significant: individuals assigned to treatment had a 2 percentage point higher likelihood of working at all, a 1.2 percentage point higher likelihood of working 20 hours or more per week, and earned 5.6% higher income. We find similar magnitude, but statistically significant, impacts on measures of the quality of employment: a 2.0 percentage point increase in formal employment, 8.6 percentage point increase in formal income, and an increase in occupational status. Our point estimates suggest the training impacts were largest for males aged above 25, even though this group was least likely to take a course conditional on being assigned to treatment. We cannot, however, reject equality of impacts by age and gender, nor do we find robust heterogeneity with respect to other individual characteristics. Consistent with these modest overall impacts, we do not find treated individuals to be in better mental health, to be any more likely to expect to be working in 2

[3] This study is also to our knowledge the first randomized experiment of any social policy in Turkey.

years' time, or to expect a higher future subjective well-being than individuals who are not trained. An expectations elicitation exercise reveals these impacts to be substantially smaller than anticipated by either program applicants or Employment Agency staff. Using administrative data to examine longer-term impacts, we confirm a small, but positive significant impact on formal employment over a one year horizon, which dissipates over time so that there is no significant impact on formal employment three years after training. As a consequence, despite the popularity of such programs globally, and their widespread usage in Turkey, our cost-benefit analysis shows that the cost of providing these programs is on average five times the benefit to participants, resulting in a strongly negative return on investment.

We then examine heterogeneity of impacts along a pre-specified causal chain of training impact in an attempt to understand why the impacts were lower than expected, and under what circumstances they are higher. We test which course characteristics associated with training quality matter, and we that find attendance rates for courses are relatively high, and there is little evidence of heterogeneity by course length, teacher quality measures, participant assessments of what the course teaches, or by the unemployment rates in the labour markets in which courses are taught. Instead, we find training to have stronger impacts when offered by private providers, with this heterogeneity being robust to adjustments for multiple hypothesis testing. Controlling for observable differences in other course characteristics, local labour market conditions and the applicants for different courses does not change this conclusion. This finding that impacts are higher with private provision is consistent with some non-experimental work (e.g. Jespersen et al, 2008) but to our knowledge this is the first time it has been found in an experimental setting. However, longer-term administrative data show that the private provider effect is no longer significant in the medium term. As a result, cost-benefit analysis suggests that even privately-run courses struggle to provide positive returns.

The remainder of the paper is structured as follows: Section 1 discusses the context and Turkey's vocational training courses, Section 2 the experimental design, and Section 3 our data collection and estimation methodology. The main results are presented in Section 4, while Section 5 traces out a causal chain from training to employment. Section 6 discusses cost-effectiveness and Section 7 concludes.

1. Context and Turkey's Vocational Training Courses

Turkey is a middle income country with a population of almost 75 million, and per capita income of US$10,400 ($17,300 in PPP terms). Urbanization, income and unemployment vary significantly at the province level. Nationwide, the population is 70% urban. In 2010, the employment rate for 15-64 year olds in Turkey was 46.3% compared to an E.U. average of 64.1% and a rate of 66.7% in the U.S.[4] This low employment rate is driven partly by a female employment rate of only 23%, but also reflects relatively high unemployment. The unemployment rate was 12.5% in 2009, and as is common in much of the world, was much higher for youth than for older workers (16.8% for youth under 25 in 2011, versus 7.2% for 25-64 year olds).[5]

The Turkish National Employment Agency (ISKUR) provides services for individuals who register as unemployed through 109 offices in 81 provinces. Training programs are the primary

[4] Source: Eurostat http://epp.eurostat.ec.europa.eu/statistics_explained/index.php/Employment_statistics [accessed December 14, 2012].

[5] Source: Eurostat http://epp.eurostat.ec.europa.eu/statistics_explained/index.php/Unemployment_statistics [accessed December 14, 2012].

active labour market program provided by ISKUR.[6] The Turkish Government dramatically increased access to ISKUR-supported training programs from 32,000 individuals trained in 2008 to 214,000 in 2010 and 250,000 in 2011. This expansion began as a means to mitigate the impact of labour market reforms and continued as a response to a spike in unemployment that coincided with the 2009 global recession.[7]

The majority of training courses offered are general vocational training courses covering a wide range of vocations[8]. These are contracted to a mix of public and licensed private providers. Courses are announced in ISKUR offices, on the ISKUR website, and by text messages. The courses are provided free to the trainees, and the trainees also receive a small stipend of 15 Turkish Lira (US$10 in 2010) per day during the course period (which averages three months). Given excess demand for courses and a desire to get the unemployed into jobs, individuals are only allowed to take one ISKUR-supported course in a five-year period. To be eligible to participate in the course, individuals must be at least 15 years in age, have at least primary education, and meet other skill pre-requisites which depend on the course they wish to participate in (for example, software courses may require some pre-existing IT knowledge or skills).

[6] ISKUR provides information about job openings to the unemployed, but does not have the intensive job search support and counseling that is often provided to the unemployed in rich countries.

[7] Note that Turkey experienced a contraction in 2009, but the economy then recovered quickly, with 9 percent growth in 2010 and 8.5 percent in 2011, and the unemployment rate dropped back to the average pre-crisis level by February 2011.

[8] ISKUR also offers training programs designed to provide convicts and ex-convicts with skills to enter the labor market after release; courses for starting a business; a small public works program; and on-the-job training programs. General vocational training accounted for 61 percent of the total number of participants in all of ISKUR's active labor market programs in 2009.

2. Experimental Design

During this period of rapid expansion of provision of vocational training services, the Turkish Ministry of Labour asked the World Bank for assistance in evaluating the impact of these courses. Excess demand among the unemployed for many of the courses offered by ISKUR provided the possibility for an oversubscription design. Moreover, it enabled evaluation of courses being offered at scale, rather than on a pilot basis.

2.1 Selection of Provinces

Our desire was to choose provinces in order to ensure a broad geographic distribution and range of labour market conditions. Selection of provinces to conduct the evaluation in began with a list of the 39 provinces which had had at least two significantly oversubscribed training courses in 2009. These provinces were first stratified by whether they had an unemployment rate above or below the median of 10% in 2009. Ten provinces were then randomly selected from each strata with probability proportional to the percentage of individuals trained in 2009. Three additional provinces (Antalya, Gaziantep, and Diyarbakir) were included in the sample at the request of ISKUR because of their importance in representing varying labour market conditions across Turkey. As a result, 23 provinces were selected for inclusion in the evaluation.[9]

2.2 Selection of Evaluation Courses

Power calculations gave a target sample size of 5,700 individuals. This target was divided amongst the 23 provinces in proportion to the number of trainees in these provinces in the previous year. Thus Istanbul accounts for 21.8% of the sample, Kocaeli, Ankara and Hatay collectively another 28%, and the remaining half of the sample is split among the other 19 provinces.

[9] The working paper (Hirschleifer et al, 2014) provides a map of the selected locations.

The evaluation team worked with regional ISKUR offices to determine the actual courses from within each province to be included in the evaluation. The key criteria used to decide which courses to include in the evaluation were i) the likelihood of the course being oversubscribed (which ensures the most popular types of training, for which there would be demand for further scale-up, are included); ii) inclusion of a diversity of types of training providers to enable comparison of private and public course provision; and iii) course starting and ending dates. The evaluation includes courses that started between October and December 2010 and finished by May 2011 (75% had finished by the end of February 2011). The timing of the evaluation was determined by the fact that it tends to be a time of year when people in Turkey are more likely to seek training through ISKUR.

This resulted in a set of 130 evaluation courses spread throughout Turkey, of which 39 were offered by private providers and the remainder were mainly government-operated. The single most common course was computerized accounting, which 24% of trainees applied for. Twenty-one percent of trainees were in service courses (babysitter, cashier, waiter, caring for the elderly), 15.4% were craftsman or machine operators (welder, natural gas fitter, plumber, mechanic), 14.7% were in technical courses (computer technicians, computer-aided design, electrical engineering), and 12.2% were in professional courses (web designers, computer programmers, IT support specialists). The average course size was 28 trainees, and the average course length was three months (typically around 6 hours per day), both with significant variation.[10] Three months is the same as the average length of the classroom components in both the Colombian and Dominican Republic training evaluations (Attanasio et al, 2011; Card et al, 2011), although those programs also supplemented this with two to three month internships to provide further on-

[10] The maximum course length in our sample is 128 days, with 90 percent of course participants in courses of 90 days or fewer.

the-job training, and is also as long as the apprenticeship program to teach vocational skills in the Malawi study (Cho et al, 2013).

2.3 Assignment of Individuals to Treatment and Control within Courses

Courses were advertised and potential trainees applied to them following standard procedures. Applications were then screened to ensure they met the eligibility criteria of ISKUR and the course provider. Training providers were then asked to select a list of potential trainees that was at least 2.2 times capacity. Typically this involved short interviews with eligible applicants. Courses in which between 1.8 and 2.2 times the course capacity were deemed suitable candidates were also included, although in those cases less than the full class size was allocated to treatment. These individuals' application details were then submitted into ISKUR's Management Information System (MIS).

The MIS system then stratified applicants for each course by gender and whether or not they were aged less than 25. Within these strata, the MIS then randomly allocated trainees at the individual level into one of three groups: a treatment group who were selected for training, a control group who were not, and a waitlisted group who the training provider could select into the training if there were drop-outs. Since training providers are paid on the basis of number actually trained, if individuals assigned to treatment drop out of training, providers look to quickly fill in the empty spots. In Card et al. (2011)'s evaluation in the Dominican Republic, this led to one-third of the control group being offered treatment, with this selection typically non-random. The inclusion of a waitlist was done to prevent this from occurring. Thus if a course had capacity for 50 trainees, and 120 were deemed eligible, 50 would be randomly assigned to treatment, 50 to control, and 20 to a waitlist.[11] If individuals from the treatment group dropped

[11] If between 1.8 and 2.2 times the course capacity applied, then proportionally smaller groups were chosen. For example, if 100 people applied for a course with capacity 50, then 50 individuals would be allocated to the training

out before one-tenth of the training course had been completed, providers could draw replacements as they liked from the waitlist. If they exhausted the first waitlist, then they drew from a second waitlist of individuals who had just missed the cut of being in the top group of applicants. At no time were trainees or prospective trainees informed of the evaluation. We do not use the waitlist in our study, since their selection into training is non-random.

The final evaluation sample consists of 5,902 applicants, of which 3,001 were allocated to treatment and 2,901 to control. There are 173 individuals who applied to more than one course. These individuals were still randomly selected into treatment or control, but have a higher probability of selection since they participate in more than one course lottery. Our estimation strategy accounts for this.

2.4 Compliance with Treatment

As is common with many training programs, not all those accepted into the course took up the training. Twenty-three percent of the individuals assigned to treatment chose either to not take training or had dropped by the second day of the course, despite it being typically only two to three weeks between the interviews for a course and its start date. There was relatively little further dropout during the course; 72% of the treatment group completed training and 69% of the treatment group received a certification of course completion in an evaluation course. Compliance does vary substantially by sub-group, with 72% of women and 62% of men receiving certification; people under 25 were also more likely to complete training. Another 3% of the treatment group as well as 3% of the control group received certification in an ISKUR

(of which 40 would be designated the treatment group for surveying and measurement purposes), 40 would be allocated to control, and 10 to the waitlist.

course that was not included in the sample of evaluation courses during the period that evaluation courses took place.[12]

3. Data Collection, Baseline Comparisons, and Estimation Methodology

The MIS system contained basic information about the course and the sex, age, and education level of the applicant. The main data for evaluation then come from surveys administered to the applicants, with some supplementary data from a survey of training providers, and some longer-term information on formal employment from the social security system.

3.1. Surveys

A baseline survey and follow-up survey were both conducted through in-person interviews by a third-party professional survey firm (Frekans) that was not affiliated by ISKUR and which was selected by the evaluation team. The baseline survey took place on a rolling basis between 13 September, 2010 and 31 January, 2011. The goal was to conduct the surveys before courses began, but given the short window of time between selection of applicants and the start of the course, in practice only one-third of those surveyed were surveyed before the start of the course, while 79% of those interviewed were interviewed with 11 days of the start of the course. Applicants were told that the purpose of the survey was to help improve the services offered by ISKUR, and that their participation in the survey had no impact on being accepted into any training course, nor would their data be shared with ISKUR at the individual level. The overall baseline response rate was 90%.

[12] An additional 7.3% of the control group and 2.7% of the treatment group took a non-ISKUR course at any time from the beginning of the study period up to the time of the baseline survey. The majority of these courses were funded by other government entities. Our power calculations accounted the possibility that at least 15% might ultimately take up training.

The follow-up survey took place between December 27, 2011, and March 5, 2012, which corresponds to a period approximately one year after the end of training. It collected data on employment outcomes, as well as individual and household well-being. The response rate was 94%, including 472 individuals who had not been able to be interviewed at baseline. In total 5,057 individuals were interviewed at both baseline and follow-up.

The attrition rate of 6% at follow-up compares favourably with the attrition rates in other evaluations of vocational training in developing countries (18.5% in Attanasio et al, 2011; 23% in Maitra and Mani, 2012; 38% in Card et al, 2011, and 46% in Cho et al, 2013). Table 1 examines whether attrition from either the baseline or follow-up surveys is related to treatment status, both for the full sample, and then by the four gender by age group stratum. To do this, we estimate the following regression:

$$Attrition_i = \alpha + \beta AssignedtoTraining + \sum_{s=1}^{457} \theta_s \delta_{i,s} + \varepsilon_i \quad (1)$$

Where *i* denotes individuals, *s* denotes a course*gender*age group lottery, and $\delta_{i,s}$ is a dummy variable indicating whether individual *i* applied for lottery *s*.[13] This controls for randomization strata (Bruhn and McKenzie, 2009) as well as controlling for individuals who applied for more than one course (Abdulkadiroglu et al, 2011). We are then interested in β, the impact of being assigned to receive vocational training, on attrition.

Table 1 reports the results. Attrition at baseline is 2.7 percentage points lower for the treatment group than the control, while attrition at follow-up is 1.4 percentage points lower. This differential attrition comes from relatively higher attrition from males in the control group than for females or for treated males, while there is no significant differential attrition for females. Whilst statistically significant, this difference in attrition rates by treatment status is small in

[13] There are 457 strata, reflecting the non-empty cells from 130 courses*2 genders*2 age groups.

absolute terms. We examine the sensitivity of our results to differential attrition by employing Lee (2009)'s bounding approach. In addition, attrition is near zero in our administrative data (only 6 individuals could not be matched). This gives us several outcome measures which are not subject to concerns about attrition. Furthermore, using this administrative data, we are unable to reject that attrition rates in the follow-up survey are unrelated to the likelihood of having found a formal job by the time of this survey (p=0.807).

3.2 Administrative Data on Employment from Social Security Records

Our sample of applicants was linked by ISKUR to official data on worker payments filed in the social security system. Due to updates and delays in this system, it took us two years to receive this data from when it was requested, and we were limited in the information that could be extracted. First, we received formal employment status, and earnings reported for social security, for the month of August 2013. This corresponds to a time period of approximately one and a half years after our follow-up survey, and two and a half years after the end of training. Secondly, in December 2013, we were given data on the first date that a worker was registered in the social security system after the end date of the course which he or she had applied for, as well as the date that they left this job, if this job ended. This data effectively covers the period from January 2011 up to the end of November 2013. It tells us month by month whether an individual was ever formally employed post-training, but since it does not report on second and subsequent formal jobs, not current formal employment status. We use this data to measure whether an individual had been formally employed by January 2012 (the mid-point of the follow-up survey), as well as to examine the trajectory of entry into formal employment.

Although we cannot fully capture the longer-term impact of training on total employment, our follow up survey indicates that, conditional on being employed at 20 hours a week, 82% the sample is formally employed. In addition, to the extent that training has any impact in measures from the follow up survey, it is concentrated in employment quality measures, such as formal employment

3.3 Balance and Baseline Characteristics

Random assignment to treatment and control occurred at the individual level and was done by computer using code written by the evaluation team. Consistent with this, the first few columns of Table 2 show balance on course and demographic characteristics using the administrative data available for the full sample. The remaining columns then examine balance for the sample interviewed at baseline, and for the sample interviewed at follow-up. Given that some baseline interviews took place after the course had started, we focus on comparing either time invariant or slow-moving individual characteristics collected during the baseline survey. The differences between treatment and control are small and magnitude, and the only significant difference at the 10% level is that the treated group surveyed at baseline is marginally less likely to have worked before. Given the number of variables tested, we view this as the result of chance, and that randomization has succeeded in providing comparable treatment and control groups. Applicants for the evaluation courses are, in general, relatively young and well-educated. The average applicant is 27 years old, and approximately 73% of them have completed high school. Sixty-one percent of the evaluation sample is female. The majority have worked previously, with 63% having worked before, and an average work experience of more than three years. Nonetheless, 37% have never worked before, and a further 20% have worked for less than one year. Baseline responses to current employment are consistent with these individuals being unemployed: only 12% have worked even one hour in the previous month, and only 2% have worked at least 20 hours per week in the past month.

A comparison of the applicants to a sample of all urban unemployed using the 2009 Labour Force survey reveals that applicants are younger, more likely to be female, have less work experience, and are better educated than the average unemployed individual. Only 30% of the unemployed are female (reflecting low labour force participation rates among women), compared to 61% of applicants; only 31% of the unemployed are youth, compared to 45% of applicants; and only 42% have completed high school, compared to 73% of applicants. These differences reflect both differences in which types of unemployed individuals are more likely to apply for training courses, as well as the supply of courses, since many courses are designed for people with at least medium levels of education. Our results are therefore informative as to the effect of training for the types of unemployed who apply for vocational training courses being offered, but need not necessarily reflect the returns to vocational training for the broader population of the unemployed.

3.4 Estimation Methodology

We can measure the intent-to-treat (ITT) effect of vocational training on a particular outcome of interest by estimating the following equation, analogous to equation (1) :

$$Outcome_i = \alpha + \beta AssignedtoTraining + \sum_{s=1}^{457} \theta_s \delta_{i,s} + \varepsilon_i \qquad (2)$$

We do not control for baseline levels of outcomes since they are close to zero and provide little variation. β is then the average effect of being selected for a vocational training course on this outcome.

We can also estimate the impact of actually completing training by replacing *AssignedtoTraining* with *CompletedTraining* in (2), and instrumenting this with treatment assignment. Under the assumption that assignment to training has no impact on outcomes for those who do not complete the course, and that there are no individuals who would take courses only if assigned to the control group, this yields the local average treatment effect (LATE). This is the impact of completing training for an individual who takes up training when they are selected in the course lottery, and does not take it up otherwise. A concern with this estimation is the possibility that simply being selected for a course may affect employment outcomes, even if individuals do not take the course or drop out after only a few days. For example, individuals (or employers) may take selection for the course as a signal of quality, which may give selected applicants more confidence approaching employers or employers a spur to hire these individuals. Showing up and finding out that you don't like the course may cause reluctant job-seekers to try harder to look for work. Due to these concerns, we focus on the ITT estimates for most of our analysis, and just report the LATE estimates for our overall employment outcomes.

The primary outcomes of interest relate to employment. We consider a variety of employment measures which aim to measure whether individuals are employed at all, as well as how much they are working, how much they earn from this work, and the quality and formality of this employment. Appendix A explains how our key variables were constructed. We pre-specified these outcomes and how they would be measured in a pre-analysis plan (Casey et al, 2012) that was archived on February 6, 2012, before any follow-up survey data was received.[14] To control further for multiple hypothesis testing among employment outcomes, we follow Kling et al. (2007) in estimating a mean treatment effect on an aggregated employment index. This first

[14] The pre-analysis plan was archived and time-stamped by the J-PAL Hypothesis Registry, and is available at http://www.povertyactionlab.org/sites/default/files/documents/ISKURIE_AnalysisPlan_v4.pdf

transforms each employment outcome by subtracting the mean and dividing by the standard deviation, and then takes an average across outcomes.

In addition to estimating the overall impact of training, we are interested in exploring the heterogeneity of impacts to help understand whether certain types of courses offer larger impacts, or certain types of people benefit more from training. Our pre-analysis plan specified dimensions of heterogeneity of interest. To estimate heterogeneity with respect to characteristic *X*, we estimate

$$Outcome_i = \alpha + \beta AssignedtoTraining + \gamma AssignedtoTraining * X + \rho X + \sum_{s=1}^{457} \theta_s \delta_{i,s} + \varepsilon_i \quad (3)$$

Recently Fink et al. (2012) have criticized randomized experiments looking for heterogeneity in treatment effects for not controlling adequately for multiple hypothesis testing. They recommend the use of the Benjamini and Hochberg (1995) approach which holds constant the false discovery rate (the expected proportion of falsely rejected null hypotheses). We use this approach to examine which dimensions of heterogeneity are robust to this concern.

4. Impacts on Employment and Well-being

We begin by looking at the overall impacts on employment for the pooled sample, and then look separately by the four age*gender strata, and for heterogeneity with respect to pre-specified human capital variables. We then examine impacts on measures of current and expected future well-being.

4.1 Overall Impacts on Employment

Table 3 reports the results of estimating equation (2) for different employment outcomes measured in our follow-up survey data. The first two columns show small and insignificant positive impacts of training on the likelihood of working at all, and of working at least 20 hours or more a week. The ITT for working at all is for a 2 percentage point impact, with a 95% confidence interval of [-0.5, +4.4] percentage points. This is thus a relatively precise zero or small effect. Lee bounds to adjust for differential attrition are also fairly narrow, consistent with the amount of differential attrition being small. The LATE estimates are for a 2.9 percentage point increase in being employed at all, and a 1.9 percentage point increase in the likelihood of working at least 20 hours per week.

We also find small and statistically insignificant impacts on weekly hours worked, income from work, and on a transform of income from work which is less sensitive to outliers. In contrast, we find statistically significant (at the 10% level) impacts on several measures of job quality: the socioeconomic occupational status of the job, being formally employed, and the income earned from formal jobs. For example, we find a 2 percentage point increase from being assigned to training in the likelihood of being formally employed, with the LATE impact being 3 percentage points. Given that 29% of the control group is formally employed at follow-up, this LATE estimate is equivalent to a 10% increase in formal sector employment.

All of these different employment outcomes show modest positive impacts.[15] Averaging them together to account for multiple testing and examine the mean impact finds a positive overall impact which is not significant at the 10% level. Moreover, the size of the impact is small, with the ITT showing only a 0.04 standard deviation improvement in employment outcomes as a result of being selected for training.

Table 4 uses the social security data to measure impacts on formal employment. Consistent with our survey data, the first column shows a statistically significant 2.6 percentage point

[15] The modest size of the treatment effect cannot be explained by trainees being more likely to pursue additional education and training. The impact of treatment on the probability of being in education or training at the time of the follow-up survey is very small, negative and marginally significant (-.014, p-value = .095)

impact on the likelihood of having found a formal job by the time of the follow-up survey. However, the next three columns show that this effect disappears over time. There is no significant effect on formal employment status or formal income by August 2013, and no impact on whether they have ever found a formal job by the end of November 2013. Figure 1 examines the trajectory of impacts in more detail, plotting the proportion of treatment and control who have ever found a formal job month by month. We see that the treatment group is more likely to have found a formal job than the control group by May 2012, and that this gap lasts for about a year, including the time of the follow-up survey, before closing again.[16] By November 2013, 66% of both groups have found a formal job at some stage during the post-training period. However, there appears to be a lot of churn in employment, with the last column of Table 4 showing that 50% of both groups have also left a formal job at some point during this period. As a result, there is no lasting impact of training on formal employment.

The point estimates of the impact of training on income in Table 3 represent a 5.8% increase in monthly income, and 8.6% increase in formal income. If we consider that the course is only 3 months in duration, then these compare favourably to a return on a full year of schooling of around 8% in Turkey (Tansel and Bodur, 2012). However, the returns to education are presumed to apply year after year in the labour market, whereas our administrative data shows the return to training is at most temporary. Moreover, in section 4.5 we will see that these returns are well below what participants and policymakers expected.

[16] The control group was about 5 percentage points more likely than the control group to take a non-ISKUR training course. If those individuals entered the labor force later, it could at least partially explain the lag in the control entering the labor force. Another possible explanation is that the impact of training interacted with changing labor market conditions.

4.2 Heterogeneity of Employment Impacts by Age and Gender

Existing experimental evaluations of vocational training programs in developing countries have focused just on youth, and Attanasio et al. (2011) find different impacts by gender, with young women benefiting more from their training in Colombia than young men. Understanding the heterogeneity of impacts by age and gender is useful for relating our results to this existing literature which has largely focused on youth, as well as for potentially helping with the targeting of training programs towards those who will benefit more. Young workers and female workers have very different labour market outcomes in general than older male workers in Turkey, with lower employment rates in the absence of training. If older male workers are likely to find work anyway, we might expect the training to have lower impacts for them than for younger workers and female workers. However, conversely if employers are reluctant to hire women for other reasons than a lack of skills, then training may have lower impacts for women than for men.

Recall that our randomization stratified by gender*age group within each course. We estimate equation (3) with X as the vector of 4 gender *age group strata and report the results in Table 5. A test of equality of treatment effects across the four subgroups enables us to determine whether there is significant heterogeneity in employment effects by age or gender. Lee bounds control for the differential attrition, which is largely only an issue for males.

The key result from Table 5 is that we cannot reject the null hypothesis of equality of employment treatment effects across the four age*gender groups. This is despite the four groups having quite different employment experiences over the year since courses ended: 62.5% of men aged 25 and older have worked in the past month, compared to 48.9% of younger men, 40.3% of younger women, and only 29.5% of women aged 25 and older. The lack of a strong employment effect of training is thus not because the labour market is so weak that no one can find jobs regardless of whether or not they are trained.

If we were to look at the four subgroups separately, only males aged 25 and older show significant treatment impacts on some employment outcomes, which appear robust to Lee bounding for differential attrition. Men in this age group assigned to training are 6.9 percentage points more likely to be working, are working 2.9 hours more per week on average, and are in a higher average occupational status (although this captures both the impacts at the extensive margin (working or not) and intensive margin (jobs taken up conditional on working). The Benjamini and Hochberg (1995) sequential adjustment approach would still show the impact on working at all and on occupational status to be significant if the false discovery rate across the four age*gender subgroups is held constant at 10%, but not significant if a 5% FDR is maintained. Thus there is evidence to support an impact for males aged over 25, but we also cannot reject that this impact is not different from the impacts for the other three age*gender subgroups.

4.3 Heterogeneity of Impacts with Respect to Other Individual Characteristics

Next we examine further whether the small overall average treatment effect is masking large heterogeneity of responses across individuals with different types of human capital. If human capital and training are complements, then we would expect larger treatment effects for individuals with higher beginning levels of human capital. In contrast, if training substitutes or compensates for other forms of human capital which individuals are lacking, then we should expect larger treatment effects for individuals with lower initial human capital levels.

We consider a wide range of pre-specified measures of human capital, including education, measures of cognitive ability (Raven test and numeracy), empowerment, personality characteristics (work centrality and tenacity), and long-term unemployment (which may indicate deteriorated skills). In addition we examine heterogeneity with respect to an individual's expectations of how much they will benefit from the course (see section 4.5), and to the presence of young children in the household (which may change an individual's cost of working).

Although these measures do help predict levels of employment, Table 6 shows there is very limited heterogeneity of treatment impacts with respect to any of them. Individuals who have previously taken a training course before, and those who are the main decision-makers about work, have larger impacts of the training according to our survey outcomes. However, the significance of neither variable survives corrections for multiple hypothesis testing, nor do we see significant impacts of these variables on the administrative measures of employment. The most statistically significant coefficient is that on tenacity, which has a negative impact on the training effect on formal employment in August 2013. This could reflect that tenacious individuals will find employment anyway, so that training substitutes for a lack of this characteristic. However, the p-value of 0.010 is not small enough to survive a correction for testing heterogeneity across so many different measures.

4.4 Impacts on Well-being

Our survey measured impacts approximately one year after the completion of training. This is similar to the timing in other experimental vocational training evaluations in developing countries, and should allow long enough for most individuals to use their training in finding a job. Nevertheless, the concern remains that perhaps some of the impacts will take time to manifest. If that is the case, we should expect to see that training leads individuals to think their future job prospects and future household well-being will be better, even if there is relatively little change in current well-being. We test this in Table 7.

While 42% of the control group is currently working at all at the time of the follow-up survey, column 1 shows that 54.1% expect to be employed in two years' time. However, there is no significant impact of treatment on this expectation. Column 2 shows that training does not improve mental health. Column 3 shows a marginally significant increase in current subjective well-being, although the impact is very small in magnitude: 0.07 steps on a 10 step-ladder, equivalent to 0.04 standard deviations. While control applicants do believe that their households will be on a higher step in five years than they are today, treatment has no additional impact on this. Treated individuals also do not have significantly higher household income, but they do have significantly higher durable asset levels. Again the magnitude of the impact is small, equivalent to 0.06 standard deviations.

Overall these results are therefore consistent with at best modest increases in employment translating into modest increases in current household wealth and subjective well-being, with no impacts on expectations about the future.[17] This accords with our finding from the administrative data that any (formal) employment impacts are not long-lasting. .

4.5 How do These Results Compare to the Expectations of Participants and Policymakers?

To address the issue of whether these impacts are in line with the expectations of both the individuals applying for ISKUR training courses and the policymakers in charge of these courses, we conducted an expectations elicitation exercise (Groh et al, 2012). Our baseline survey elicited subjective probabilistic expectations (Delavande et al, 2010) by asking

[17] Our pre-analysis plan also hypothesized that training may further impact on social outcomes such as whether individuals head their own household, their decision-making power, and their attitudes towards women's role in society. We find no significant effects on any of these measures, which is consistent with the lack of strong labor market impacts (results available upon request).

participants what they thought was the percent chance they would be employed in one year if they were selected for ISKUR training, and if they were not. A sample of 51 ISKUR employees from across the different provinces were also asked a range question about the likelihood the control group would be employed, along with a question eliciting the percent difference in employment rates they would expect from training.

Table 8 reports the results. The first two columns show that on average individuals were reasonably close in terms of their expectations of what employment levels would be like in the absence of training: the mean response was for a 31% chance of being employed, and in practice 36% of the control group is employed. Males aged 25 and over seem to underestimate their higher likelihood of being employed than the other groups, while older females do have lower expectations of being employed to match their lower realized levels.

In contrast to these relatively accurate expectations of the likelihoods of being employed without training, individuals dramatically overestimate the benefits from training: our LATE estimate is for a 2 percentage point increase in employment as a result of training, whereas the mean expected increase among those assigned to the treatment group is for a 32.4 percentage point increase.[18] This overestimation occurs for all gender and age groups. Moreover, ISKUR staff also overestimated the gain from training, with a mean expected treatment gain of 24.3 percentage points. The impact of training is thus much less than both the staff at ISKUR and the training applicants anticipated.

Our finding that policymakers and participants dramatically over-estimate the returns to the course complements recent work by Smith et al. (2011), who examine the accuracy of program

[18] We report these expectations for the treatment group, to offset any concerns that individuals in the control group who had found out they were not selected for training by the time of the interview might understate their expectations of the value of training. In practice, however, the control and treatment group have very similar expectations, suggesting this is not a factor: the mean treatment gain expected by the control group is 31.9 percentage points, which is similar in magnitude to the 32.4 percentage point gain expected by the treatment group.

participants' evaluations of a job training program *after* taking part in such training. They find participants do not have accurate evaluations of the impact of such courses even ex post.

5. Why Does Training Have Such Limited Effects, and Do Certain Types of Training Work Better?

The impact of training is less than expected by either training participants or the labour ministry staff, although the impacts on employment are not that different from those found in other evaluations. In order for training to be effective, our pre-analysis plan set out four intermediate steps in a causal chain through which we might expect to see selection into a course influencing employment outcomes. We examine each in turn, which also helps understand whether certain types of training have more impact.[19]

5.1 Individuals Selected For Courses Must Show Up and Complete Training

The first step in our causal chain is that individuals selected for training classes must actually show up and attend these classes. As noted above, 77% of those selected for treatment attended the course beyond the second day, and 72% completed it. The LATE estimates show impacts which adjust for attendance. However, we might also think that lower attendance and completion rates are an indicator of courses that participants find to be of lower quality or expect to be less useful. We therefore estimate equation (3) by interacting treatment with the percent of individuals assigned to a course who attended the course or who completed it. Since this characteristic varies only at the course level, we cluster standard errors by course (and also do this for all subsequent tests of interactions of treatment with course characteristics).

[19] We specified that we would look at these effects for two survey outcomes: being employed for 20+ hours in a week, and the aggregate employment index. Given that we find impacts vary over time, we also examine this heterogeneity for two measures of formal employment from our administrative data: ever having a formal job between course end and January 2012 (around the follow-up survey), and being formally employed as of August 2013.

The first two rows of Table 9 show these treatment interactions. The interactions are positive, as hypothesized, suggesting that treatment effects are larger in courses with higher attendance rates. However, the results are almost all not statistically significant, and the magnitude of the impacts is not large: a one standard deviation increase in course attendance rates is associated with a 1 percentage point increase in employment. The administrative data shows a significant longer-term impact at the 10% level of the course having higher attendance, but the effect is still small in magnitude (a 1 s.d. increase is associated with a 2 percentage point increase in formal employment in August 2013), and is no longer significant after correcting for multiple hypothesis testing.

5.2 Higher Quality Courses Should Have More Impact

A second concern is whether courses averaging three months in length are long enough to teach the skills necessary to improve labour market performance. The median course length is 320 hours which does suggest enough hours to enable learning to take place. To investigate whether longer courses have bigger impacts, we interact whether or not the course is above median length with treatment. The third row of Table 9 shows that longer courses in fact have less impact on employment than shorter courses amongst the sample of courses offered here. One possible explanation for this is that individuals reduce job search whilst taking part in training, so individuals in longer training courses have had less time to look for jobs (Rosholm and Skipper, 2009). The point estimates are significant at the 10% level, but are not significant after controlling for multiple hypothesis testing. They are certainly not consistent with the view that the limited impacts are due to the courses being too short, and the long term results from administrative data also do not show better impact from longer courses.

As proxies for the qualities of the teachers, we examine in the next two rows of Table 9 the treatment interactions with whether the average experience of teachers in the course is above 12 months (the median), and with the percentage of teachers for the course who are tertiary educated. We find very small and statistically insignificant impacts of either measure. This is consistent with much of the work in the education literature, and likely reflects the finding there that education and experience explain little of the actual variation in teacher effectiveness (Hanushek and Rivkin, 2006). It does suggest that the reason for limited impacts is not that the courses are taught by staff that are not experienced or educated enough.

Previous non-experimental literature in developed countries has found some evidence to suggest that the impacts of training are higher for training offered by private providers (Jespersen et al, 2008). Possible reasons are that private training providers are more responsive to private sector employer demand and/or they potentially face more competition and thus must increase quality in response. Rows six and seven of Table 9 examine the treatment interaction with facing two or more competitors, and with being a private provider. We see positive, but not statistically significant, impacts of facing some competition. Private provision has stronger impacts, with a positive impact on employment and on having had a formal job by the time of survey which are significant at the 10% level, and a positive impact on the aggregate employment index which is significant at the 5% level. The latter survives adjustments for multiple hypothesis testing within the 5 outcomes used to proxy for course quality. Training results in a 4 to 6 percentage point higher increase in employment after 1 year in private courses than in public courses. This gap narrows to 2.5 percentage points after 3 years, and is no longer statistically significant.

Appendix Table 1 compares characteristics of private and public courses, and shows that private courses are slightly longer than publicly provided courses, and are much more likely to be accounting courses and less likely to be craftsman, technical or service courses. Applicants are more educated for private courses than public courses, but are also slightly younger and have less work experience. In the working paper version (Hirshleifer et 2014) we show this private provider short-term effect is robust to controlling for these other observable characteristics.

5.3 Skill Acquisition, Signalling, or Job Matching

A third step in the causal chain is for individuals who take courses of sufficient quality to be able to use what they have learned in the course to find jobs. There are three main channels through which vocational education may help in this respect. First, it might increase human capital through teaching new technical skills. Second, it may act to certify skills that individuals already have and act as a signalling mechanism to employers. Third, it may teach individuals new strategies for finding jobs in a certain profession, or better alert them to job opportunities, thereby improving job matching.

To examine the extent to which courses are playing each of these roles, and to which treatment effects vary with them, our follow-up survey asked course participants whether they thought the course had done each of these three things. We interact the percentage of course participants who think that the course taught new technical skills, certified existing skills, taught new strategies for finding jobs, or made them more aware of job opportunities with treatment and show the results in Table 9. We see that on average 84% of participants thought the courses certified skills they already had, and 80% thought they taught new skills, while 60% thought the course helped with job finding strategies and 45% with making them more aware of job opportunities.[20] However, we find the point estimates typically to be negative and not statistically significant. Thinking the course made them aware of new jobs has a significant

[20] Although see Smith et al. (2011) for evidence that suggests that program participants are not always good judges of whether courses they have taken have helped them.

negative interaction for the outcome of having had a formal job at the time of the follow-up survey, but this is not significant after correcting for multiple testing. Thus the heterogeneity in how participants perceive these factors across courses does not seem to drive heterogeneity in treatment outcomes.

5.4 Training Impacts and Unemployment Rates

The last step specified in the causal chain is for individuals who receive training to find jobs that they would not otherwise get. This depends on the labour market they face. On one hand, when unemployment rates are low there should be more job opportunities available, making it easier for people to use their training to find jobs. But in such cases firms facing labour shortages might hire workers regardless of whether or not they have been trained. In contrast, when unemployment rates are high, if employers are not hiring then having new skills may not help the unemployed find jobs, but on the other hand, employers may be choosier and so training may offer workers a way to distinguish themselves from other workers competing for the same jobs.

As a result, theoretically it is ambiguous whether we should expect training to have more or less impact in situations of higher or lower unemployment. Consistent with this, there are mixed results in the existing (non-experimental) literature. Lechner and Wunsch (2009) find German training programs to be more effective when carried out during times when unemployment rates are higher and Kluve (2010) finds in his meta-analysis that program effects tend to be higher when unemployment rates are higher. But using Norwegian data, Raaum et al. (2003) find training impacts to be positively correlated with post-training employment rates in local labour markets.

The training programs in our study took place throughout 23 provinces with a variety of unemployment conditions – at the time of training, unemployment rates ranged from under 5%

to over 20%, with a median of approximately 14%. The third to last row of Table 9 interacts the treatment with an indicator for above median provincial unemployment. The point estimates for our survey outcomes are positive, suggesting larger treatment impacts when labour markets are tighter, but not statistically significant. The longer-term impact is statistically significant, suggesting training to increase employment by 4.9 percentage points more when done in high unemployment regions compared to low unemployment regions, although this result does not survive a multiple hypothesis test correction.

Finally we examine heterogeneity with respect to the type of course. There is some evidence that accounting courses, the most common course type, have larger impacts on short-term employment outcomes than other types of courses. This impact is however not statistically significant in terms of formal employment, or long-term formal employment.

6. Cost-Benefit

The mean cost of a course is 1574 Turkish Lira (TL) per person, with privately provided courses having a mean cost of 1792 TL per person compared to 1455 TL per person for publicly provided courses. In addition to this, participants receive a stipend of 15 TL per day, which for an average course length of 57 days equates to a further 855 TL. The total cost to the government of providing a course thus averages 2429 TL per person (US$1619), and is 2692 TL (US$1795) per person for private courses.

Our LATE estimate of the overall gain in monthly income in Table 3 is 26 TL (although this is not statistically significant). If the course led to a sustained level increase in income of this amount, it would take 93 months for the gain in income to offset the cost of the course. However, our administrative data suggests any impacts on formal employment last at most 1.5 years, so that the costs are approximately five times the benefits, and the overall return to the course is strongly negative.

Training was more effective for privately provided courses. The LATE estimate of the overall gain in monthly income for this group is a statistically significant 66 TL per month. It would take 41 months of these gains for the benefits to exceed the costs. However, since Table 9 shows that the formal employment impact of the course is no longer significant after 2.5 years, it also seems unlikely that the benefits to private courses exceed their costs.[21] As with all other vocational training evaluations in developing countries which we are aware of, we are unable to measure any general equilibrium spillover effects. Such effects have been detected by Crépon et al. (2013) for a job placement program for youth in France, who find some of the gains to treated youth come at the expense of the untreated. To the extent that trained individuals crowd out non-trained individuals from jobs, we are understating the cost per job created and overstating the returns. Turkey has approximately 2.5 million unemployed people. The 3,000 individuals trained in this experiment are thus a trivial fraction of all the unemployed, and even the 210,000 individuals trained in total in 2010 constitute less than 10% of the unemployed. This makes it seem likely that any such spillovers or crowding out may be second-order effects in our context.

7. Conclusions

This paper provides the first randomized evaluation of vocational training programs offered to the general unemployed population in a developing country setting. A large sample, containing both youth and older unemployed, as well as courses offered by the private and the public sector,

[21] These calculations are simplifications which ignore any opportunity cost of time or lost wages for participants during the time they take the course (since few were employed at baseline such costs are low); ignore any additional costs of distortions in the economy from raising the taxes to pay for these courses; and ignore any gains in utility associated with the non-wage benefits of being in more formal or higher quality jobs. These factors seem likely to be of second-order importance in our context.

enables us to examine the effectiveness of such training for a wider range of demographic characteristics and course types than existing literature. Linking participants to social security data enables us to trace the long-term trajectory of impacts on formal employment as well as avoid the attrition and measurement concerns that face existing studies in developing countries.

Our results show that the average impact of these vocational training programs is a very modest positive overall impact on employment and on the quality of employment, with the measured impact of the courses much less than expected by either course participants or government labor ministry staff. Despite the overall impact being close to zero, we find stronger and statistically significant impacts of vocational training in courses offered by private providers. Being selected for one of these courses results in a 4 to 6 percentage point increase in employment rates, relative to an employment rate of 37% for the control group. These returns persist when we control for observable differences in the characteristics of the courses and of their participants, but do not appear to last when it comes to impacts on formal employment over a 2.5 to 3 year period. Taken together the results suggest that there is some potential for vocational training to improve the short-term employment prospects of the unemployed, but that this potential will be best realized when courses are offered by providers that have both the incentives and ability to respond to market demands. However, overall the results suggest that this large-scale vocational training program does not meet a cost-benefit test. Given the renewed emphasis of these types of programs for many governments around the world, these results suggest policymakers should be cautious in expecting such programs to have large impacts.

Appendix A: Definitions of Key Variables

Employment Outcomes

The following employment outcomes are defined based on data collected in the follow-up survey.

Working at all: An indicator variable which takes value one if the individual has worked for cash or in-kind income at all in the past four weeks.

Employed 20 hours +: An indicator variable which takes value one if the individual is currently working for 20 hours a week or more.

Weekly hours: Hours worked per week in the last month employed. This is coded at 0 for individuals currently not working, and top-coded at 100 hours per week (the 99th percentile of the baseline response) to reduce the influence of outliers.

Monthly income: Total monthly income from work in the last month. This is coded as 0 for individuals not working, and top-coded at the 99th percentile of the control group earnings distribution (2500 Turkish Lira) to reduce the influence of outliers.

Transformed monthly income: The inverse hyperbolic sine transformation of monthly income from work in the last month, $log\left(y + (y^2 + 1)^{1/2}\right)$. This is intended to be more robust to outliers than levels of income and is similar to the logarithm transformation, but is also defined when income is zero (Burbidge et al, 1988).

Occupational status: this is coded based on work occupation using the international measures of socioeconomic occupational status of Ganzeboom and Treiman (1996). This is a continuous measure ranging from 16 (e.g. domestic helpers) to 90 (e.g. judges), and is coded as zero for individuals not working.

Formal work: this is an indicator variable coded as one if the individual is currently working in a job covered by social security.

Formal income: this is monthly income earned in jobs covered by social security.

Aggregate employment index: A standardized index obtained as the average of each of the above variables, after each has been standardized by subtracting the mean and dividing by its standard deviation. This measure is set to missing for individuals who are missing data for the working at all variable, and otherwise is the average over all employment variables with non-missing data.

Ever formally employed between course end and January 2012: this is an indicator variable coded as one if the individual is reported in the social security administrative data as being in formal employment at any time between the end of their course date and January 2012, the midpoint of the follow-up survey.

Formally employed in August 2013: An indicator variable which takes the value one if social security administrative data reveal the individual to be formally employed in the month of August 2013.

Formal income earned in August 2013: Income for the month of August 2013, as reported in the social security administrative data.

Ever formally employed between course end and November 2013: : this is an indicator variable coded as one if the individual is reported in the social security administrative data as being in formal employment at any time between the end of their course date and November 2013.

Ever left a formal job between course end and November 2013: an indicator variable which is coded as one if the individual is reported in the social security administrative data as having left their first formal job worked in during this period.

Well-being Measures

Expected probability of working in two years: The expected chance of having a job in two years time, coded as missing if an answer outside of the 0 to 100 range is given.

Mental health index MHI-5: this is a five item index of Veit and Ware (1983) which has a maximum score of 25 and minimum score of 5. Higher scores are desirable in that they indicate the experience of psychological well-being and the absence of psychological distress. Individuals are asked how often in the past four weeks they have done each of the following, each answered on a 5 point scale, where 1 denotes none of the time and 5 all of the time, and the MHI-5 is the sum of these responses:

- Been a nervous person (reverse-coded)
- Felt so down in the dumps that nothing could cheer them up (reverse-coded)
- Felt calm and peaceful
- Felt downhearted and blue (reverse-coded)
- Been a happy person

Current subjective well-being: individuals are asked which step on a 10-step Cantril ladder, where on the first step stand the poorest people, and on the tenth step standard the richest, they think their household stands today.

Subjective well-being in five years: Which step on the Cantril ladder they think their household will be on in 5 years time.

Total household income in past year: Income from all sources, top coded at the 99th percentile of the control group distribution(74,000 Turkish Lira)

Transformed household income: The inverse hyperbolic sine of total household income.

Durable asset index: the first principal component of 15 indicators of household durable asset and infrastructure ownership (own a gas or electric oven; own a microwave oven; own a dishwasher; own a DVD/VCD player; own a camera; have Digiturk/Satelite; own an air conditioner; own a CD player or iPod; own a telephone; own a computer; have an internet connection; own a private car; own a taxi, minibus or commercial vehicle; own a bicycle; have 4 or more rooms in their house).

Human Capital Measures

Is the main decision maker for where they work: based on a question which asks them who in the household makes decisions over whether they can work outside the home.

Raven test score: score out of 12 on a Raven Progressive Matrices test

Numerate: was able to answer four computational questions involving time, percentages, division and subtraction correctly.

Work centrality: Answer to the question "The most important thing that happens in life involves work", where 5 = strongly agree, and 1 = strongly disagree

Tenacity: this measures the extent to which individuals persist in difficult circumstances and is taken as the sum of responses on two questions taken from Baum and Locke (2004).

Author Affiliations

Sarojini Hirshleifer, *University of California, San Diego*

David McKenzie, *World Bank*

Rita Almeida, *World Bank*

Cristobal Ridao-Cano, *World Bank*

References

Abdulkadiroglu, A., Angrist, J., Dynarski, S., Kane, T. and Pathak, P. (2011). “Accountability and flexibility in public schools: Evidence from Boston’s charters and pilots.” *Quarterly Journal of Economics* vol. 126(2), pp. 699-748.

Attanasio, O., Kugler, A., Meghir, C. (2011). “Subsidizing vocational training for disadvantaged youth in Colombia: Evidence from a randomized trial”, *American Economic Journal: Applied Economics* vol. 3(3), pp. 188-220.

Bandiera, O., Buehren, N., Burgess, R., Goldstein, M., Gulesci, S., Rasul, I., and Sulaiman, M. (2012). “Empowering adolescent girls: Evidence from a randomized control trial in Uganda”, Mimeo. LSE.

Baum, J.R. and Locke, E.A. (2004). “The relationship of entrepreneurial traits, skill, and motivation to subsequent venture growth”, *Journal of Applied Psychology* vol. 89(4), pp. 587-98.

Benjamini, Y. and Hochberg, Y. (1995). “Controlling the false discovery rate: A practical and powerful approach to multiple testing”, *Journal of the Royal Statistical Society Series B*, vol. 57(1), pp. 289-300.

Betcherman, G., Olivas, K., and Dar, A. (2004) “Impacts of active labor market programs: New evidence from evaluations with particular attention to developing and transition countries.” World Bank Social Protection Discussion Paper Series 0402.

Blattman, C., Fiala, N., and Martinez, S. (2014). “Generating skilled self-employment in developing countries: Experimental evidence from Uganda”, *Quarterly Journal of Economics*, vol. 129(2), pp. 697-752.

Bruhn, M. and McKenzie, D. (2009), “In pursuit of balance: Randomization in practice in development field experiments,” *American Economic Journal: Applied Economics* vol. 1(4), pp. 200-232.

Burbidge, J., Magee, L. and Robb, A.L. (1988). "Alternative transformations to handle extreme values of the dependent variable" *Journal of the American Statistical Association* vol. 83(401), pp. 123-127

Card, D., Ibarraran, P., Regalia, F., Rosas-Shady, D., and Soares, Y. (2011) “The labor market impacts of youth training in the Dominican Republic”, *Journal of Labor Economics* vol. 29(2), pp. 267-300.

Casey, K., Glennerster, R. and Miguel, E. (2012) “Reshaping institutions: Evidence on aid impacts using a pre-analysis plan”, *Quarterly Journal of Economics* vol. 127(4), pp. 1755-1812.

Cho, Y., Kalomba, D., Mobarak, A.M. and Orozco, V. (2013). "Gender differences in the effects of vocational training: Constraints on women and drop-out behavior", *IZA Discussion Paper no. 7408.*

Crépon, B., Duflo, E., Gurgand, M., Rathelot, R., and Zamora, P. (2013). "Do labor market policies have displacement effects? Evidence from a clustered randomized experiment", *Quarterly Journal of Economics* vol. 128(2), pp. 531-80.

Delavande, A., Gine, X. and McKenzie, D. (2010) "Measuring subjective expectations in developing countries: A critical review and new evidence", *Journal of Development Economics*, vol. 94(2), pp. 151-63.

Fink, G., McConnell, M. and Vollmer, S. (2012). "Testing for heterogeneous treatment effects in experimental data: False discovery risks and correction procedures", Mimeo. Harvard School of Public Health.

Ganzeboom, H. and Treiman, D. (1996). "Internationally comparable measures of occupational status for the 1988 International Standard Classification of Occupations", *Social Science Research* vol. 25(1), pp. 201-39.

Groh, M., Krishnan, N., McKenzie, D. and Vishwanath, T. (2012). "Soft skills or hard cash? The impact of training and wage subsidy programs on female youth employment in Jordan", *World Bank Policy Research Working Paper* no. 6141.

Hanushek, E. and Ritkin, S. (2006). "Teacher Quality", pp. 1051-78 in E. Hanushek and F. Welch (eds.) *Handbook of the Economics of Education Volume 2*. Elsevier, Amsterdam.

Heckman, J.J., Lalonde, R.J. and Smith, J.A. (1999). "The economics and econometrics of active labor market programs", pp. 1865-2097 in *Handbook of Labor Economics Volume 3, Part A*, Elsevier, Amsterdam

Hirshleifer, S., McKenzie, D., Almeida, R., and Ridao-Cano, C. (2014). “The impact of vocational training for the unemployed: Experimental evidence from Turkey”, *World Bank Policy Research Working Paper no.* 6807,

International Labour Organization (ILO) (2012) *World of work report 2012: Better jobs for a better economy*. ILO, Geneva.

Jespersen, S., Munch, J. and Skipper, L. (2008). “Costs and benefits of Danish active labour market programmes”, *Labour Economics* vol. 15(5), pp. 859-84.

King, K. and Palmer, R. (2010). *Planning for technical and vocational skills development*. UNESCO: International Institute for Educational Planning, Paris.

Kling, J.R.., Liebman, J.B. and Katz, L.F. (2007). ‘‘Experimental analysis of neighborhood effects,’’ *Econometrica*, vol. 75 (1), pp. 83–119.

Kluve, J. (2010). “The effectiveness of European active labour market policy”, *Labour Economics* vol. 17(6), pp. 904-18.

Lechner, M. and Wunsch, C. (2009) “Are training programs more effective when unemployment is high?”, *Journal of Labor Economics* vol. 27(4), pp. 653-92.

Lee, D. (2009). “Training, wages, and sample selection: Estimating sharp bounds on treatment effects", *Review of Economic Studies* vol. 76(3), pp. 1071-1102.

Maitra, P. and Mani, S. (2012). “Learning and earning: Evidence from a randomized evaluation in India”, Mimeo. Department of Economics, Monash University.

McKenzie, D. and Robalino, D. (2010) “Jobs and the crisis: What has been done and where to go from here?”, *Viewpoint* note number 325, The World Bank Finance and Private Sector Development Vice Presidency.

Raaum, O., Torp, H. and Zhang, T. (2003) “Business cycles and the impact of labour market programmes”, Oslo University Department of Economics working paper 14/2002

Rosholm, M. and Skipper, L. (2009). “Is labour market training a curse for the unemployed? Evidence from a social experiment”, *Journal of Applied Econometrics*, vol. 24 (2), pp. 338-365

Schochet, P., Burghardt, J. and McConnell, S. (2008). “Does Jobs Corp work? Impact findings from the national Job Corps study”, *American Economic Review* vol. 98(5), pp. 1864-86.

Smith, J., Whalley, A. and Wilcox, N. (2011) “Are program participants good evaluators?”, Mimeo. University of Michigan. http://www-personal.umich.edu/~econjeff/Papers/SWW%20JTPA%20Participant%20Evaluation%20Full%20032111.pdf (last accessed: July 25, 2014).

Tansel, A. and Bodur, F.B. (2012). “Wage inequality and returns to education in Turkey: A quantile regression analysis”, *Review of Development Economics* vol. 16(1), pp. 107-21.

UNESCO (2012) *Education for all Global Monitoring Report 2012: Youth and Skills – putting education to work.* UNESCO, Paris.

World Bank (2012a) *Skills, Not Just Diplomas: Managing Education for Results in Europe and Central Asia.* World Bank, Washington D.C.

World Bank (2012b) *World Development Report 2013: Jobs*. World Bank, Washington D.C.

Figure 1: Trajectory of Formal Employment Impact

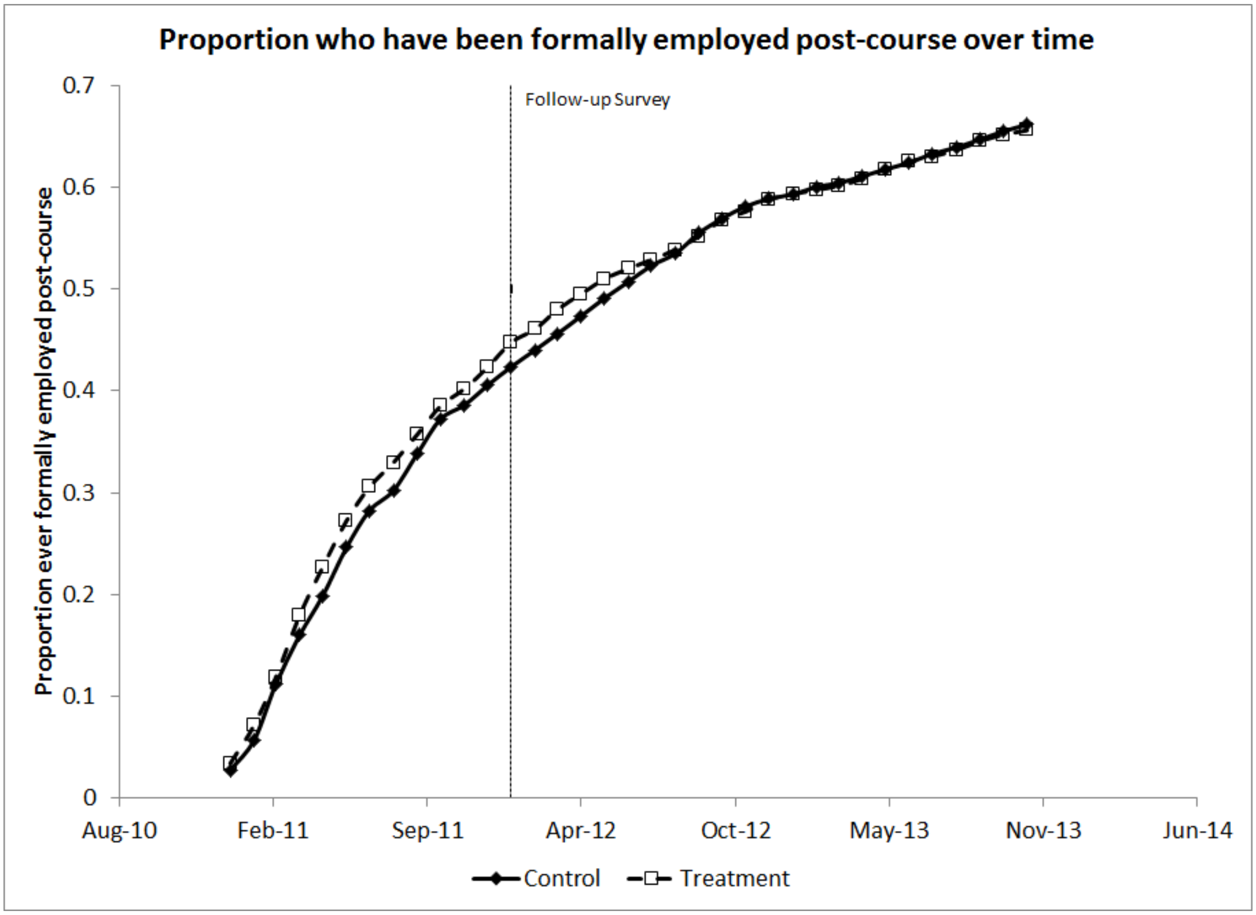

Table 1: Is Attrition Related to Treatment Status?

	Full Sample		Male Youth		Older Males		Female Youth		Older Females	
	Baseline	Follow-up	Baseline	Follow-up	Baseline	Follow-up	Baseline	Follow-up	Baseline	Follow-up
Assigned to Treatment	-0.027***	-0.014**	-0.059***	-0.031**	-0.059***	-0.038**	-0.020	-0.001	0.001	-0.002
	(0.008)	(0.006)	(0.020)	(0.015)	(0.018)	(0.015)	(0.015)	(0.012)	(0.012)	(0.010)
Control group attrition rate	0.114	0.070	0.155	0.085	0.130	0.084	0.111	0.064	0.087	0.058
Sample size	5902	5902	1146	1146	1059	1059	1583	1583	2114	2114

Notes: Robust standard errors in parentheses. *, **, and *** indicate significance at the 10, 5 and 1% levels respectively.
All regressions include controls for course*gender*age group lotteries applied for.
Youth denotes individuals under 25, Older denotes individuals 25 and up.

Table 2: Balance and Summary Statistics

	Full Experimental Sample			Baseline Sample			Follow-up Sample		
	N	Control Mean (S.D.)	Treatment Difference (Std. Error)	N	Control Mean (S.D.)	Treatment Difference (Std. Error)	N	Control Mean (S.D.)	Treatment Difference (Std. Error)
Course Characteristics									
Course length (days)	5877	56.5	-0.005	5284	56.4	0.038	5507	56.4	-0.044
		(23.4)	(0.073)		(23.2)	(0.082)		(23.4)	(0.071)
Course length (hours)	5877	336	0.066	5284	334	0.246	5507	335	-0.241
		(151)	(0.453)		(150)	(0.506)		(151)	(0.439)
Private Provider	5899	0.349	0.000	5305	0.347	0.001	5526	0.351	0.000
		(0.477)	(0.002)		(0.476)	(0.002)		(0.477)	(0.002)
Accounting course	5877	0.242	0.000	5284	0.240	-0.000	5507	0.244	0.000
		(0.428)	(0.002)		(0.427)	(0.002)		(0.429)	(0.002)
Professional course	5877	0.122	-0.001	5284	0.116	-0.000	5507	0.120	-0.001
		(0.328)	(0.001)		(0.320)	(0.001)		(0.325)	(0.001)
Craftsman course	5877	0.156	-0.000	5284	0.155	-0.000	5507	0.158	-0.000
		(0.363)	(0.001)		(0.362)	(0.001)		(0.364)	(0.001)
Technical course	5877	0.147	0.000	5284	0.148	-0.000	5507	0.144	0.001
		(0.354)	(0.001)		(0.356)	(0.001)		(0.351)	(0.001)
Service course	5877	0.211	0.001	5284	0.217	0.000	5507	0.211	0.000
		(0.408)	(0.001)		(0.413)	(0.001)		(0.408)	(0.001)
Course in Istanbul	5902	0.220	-0.000	5308	0.217	-0.001	5529	0.211	0.000
		(0.414)	(0.001)		(0.412)	(0.001)		(0.408)	(0.001)
Individual Characteristics									

(administrative data)									
Female	5902	0.617	-0.000	5308	0.629	-0.002	5529	0.623	-0.001
		(0.486)	(0.002)		(0.483)	(0.002)		(0.485)	(0.002)
Age	5902	27.0	-0.041	5308	27.1	-0.089	5529	27.0	-0.058
		(7.2)	(0.109)		(7.2)	(0.116)		(7.2)	(0.113)
At least high school	5902	0.725	-0.004	5308	0.724	-0.005	5529	0.724	-0.004
		(0.447)	(0.009)		(0.447)	(0.009)		(0.447)	(0.009)
Individual characteristics (baseline data)									
Years of education				5255	11.3	-0.005	5008	11.3	0.014
					(3.3)	(0.069)		(3.3)	(0.071)
Has done previous training course				5308	0.264	-0.007	5057	0.265	-0.008
					(0.441)	(0.012)		(0.441)	(0.012)
Household head				5276	0.134	-0.000	5027	0.133	-0.003
					(0.340)	(0.008)		(0.340)	(0.008)
Household size				5308	4.09	0.024	5057	4.10	0.011
					(1.57)	(0.040)		(1.57)	(0.040)
Married				5033	0.346	0.004	4797	0.351	0.002
					(0.476)	(0.011)		(0.478)	(0.012)
Ever employed				5308	0.631	-0.021*	5057	0.626	-0.020
					(0.483)	(0.012)		(0.484)	(0.013)
Total years working for pay				5277	3.38	0.006	5027	3.33	0.006
					(4.91)	(0.111)		(4.88)	(0.113)

Note: Standard errors are robust standard errors. *, **, and *** indicate significance at the 10, 5 and 1% levels
respectively. Standard errors based on regression with controls for course*gender*age group lotteries applied for.
Sample sizes for the follow-up data for the individual characteristics measured at baseline are for the sample interviewed
at both baseline and follow-up - there are also individuals interviewed at follow-up but not

at baseline.

Table 3: Impact of Training on Employment Outcomes (Survey Data)

	Working at all	Employed 20 hours+	Weekly Hours	Monthly Income	Transformed Monthly Income	Occupational Status	Formal Work	Formal Income	Aggregate Employment Index
ITT Estimate	0.020	0.012	0.860	17.316	0.121	0.962*	0.020*	22.166*	0.036
	(0.013)	(0.013)	(0.680)	(12.271)	(0.093)	(0.573)	(0.012)	(11.965)	(0.024)
Lee lower bound ITT	0.011	0.003	-0.103	-4.959	0.050	0.922	0.011	-0.231	0.012
	(0.013)	(0.013)	(0.668)	(11.720)	(0.093)	(0.573)	(0.012)	(11.378)	(0.024)
Lee upper bound ITT	0.026**	0.018	1.145*	22.592*	0.165*	1.242**	0.025**	26.460**	0.939***
	(0.013)	(0.013)	(0.684)	(12.331)	(0.094)	(0.576)	(0.012)	(12.034)	(0.027)
LATE Estimate	0.029	0.019	1.294	25.989	0.182	1.452*	0.030*	33.243*	0.054
	(0.018)	(0.018)	(0.980)	(17.624)	(0.134)	(0.828)	(0.017)	(17.184)	(0.035)
Control Mean	0.420	0.361	17.922	299.109	2.541	17.128	0.293	257.887	0.007
Control Standard Deviation	0.494	0.480	25.545	464.600	3.516	21.763	0.455	448.160	0.936
Sample Size	5497	5529	5439	5396	5396	5418	5508	5464	5497

Notes: Robust standard errors in parentheses. *, **, and *** indicate significance at the 10, 5 and 1% levels respectively.
All regressions include controls for course*gender*age group lotteries applied for.
Appendix A defines the outcome variables.

Table 4: Impact of Training on Employment as Measured in Social Security Data

	Ever formally employed between course end and Jan 2012 (time of follow-up survey)	Formally employed in August 2013	Formal Income earned in August 2013	Ever formally employed between course end and November 2013	Ever left a formal job between course end and Nov 2013
ITT Estimate	0.026**	-0.009	-4.427	-0.005	0.015
	(0.012)	(0.012)	(19.935)	(0.011)	(0.012)
Control Mean	0.424	0.427	553.820	0.662	0.506
Control Std. Dev.	0.494	0.495	796.571	0.473	0.500
Number of Observations	5896	5896	5896	5896	5896

Notes: Robust standard errors in parentheses. *, **, and *** indicate significance at the 10, 5 and 1% levels respectively.
All regressions include controls for course*gender*age group lotteries applied for.

*Table 5: Heterogeneity in Employment Outcomes by Age*Gender Stratifying Variables*

	Working at all	Employed 20 hours+	Weekly Hours	Monthly Income	Transformed Monthly Income	Occupational Status	Formal Work	Formal Income
ITT for males under 25	-0.021	-0.034	-0.952	-12.112	-0.241	-0.495	0.003	20.734
	(0.030)	(0.030)	(1.626)	(31.384)	(0.223)	(1.299)	(0.028)	(30.160)
	[-0.04, -0.01]	[-0.06*, -0.02]	[-2.5, -0.3]	[-58**, 1]	[-0.40*, -0.15]	[-1.4, +0.1]	[-0.02, 0.02]	[-24.9, 33.0]
ITT for males over 25	0.069** (a)	0.047	2.917*	35.304	0.383	2.997**(a)	0.047	32.019
	(0.029)	(0.031)	(1.772)	(37.501)	(0.244)	(1.332)	(0.032)	(36.733)
	[0.06*, 0.10***]	[0.03, 0.07**]	[1.0, 4.2**]	[-17, 62*]	[0.23, 0.58**]	[1.4, 4.3***]	[0.03, 0.07**]	[-28, +55]
ITT for females	0.011	0.016	0.952	14.938	0.140	0.567	0.017	15.571

under 25								
	(0.025)	(0.025)	(1.355)	(20.999)	(0.181)	(1.161)	(0.024)	(20.519)
	[0.01, 0.01]	[0.02, 0.02]	[0.8, 1.0]	[12, 15]	[0.13, 0.14]	[0.5, 0.6]	[0.02, 0.02]	[13, 16]
ITT for females over 25	0.023	0.018	0.761	26.052	0.174	1.034	0.019	23.058
	(0.020)	(0.019)	(1.001)	(17.321)	(0.141)	(0.917)	(0.018)	(16.788)
	[0.02, 0.03]	[0.02, 0.02]	[0.7, 0.8]	[22, 27]	[0.16, 0.18]	[1.0, 1.1]	[0.02, 0.02]	[19, 24]
p-value for testing equality:	0.188	0.276	0.457	0.712	0.268	0.294	0.768	0.982
Control Mean: young men	0.489	0.407	20.1	347	2.90	19.2	0.307	275
Control Mean: older men	0.621	0.548	27.3	518	3.94	24.3	0.466	458
Control Mean: young women	0.403	0.356	18.1	269	2.50	17.3	0.295	238
Control Mean: older women	0.295	0.247	12.0	187	1.69	12.4	0.196	161
Sample size	5497	5529	5439	5396	5396	5418	5508	5464

Notes: Robust standard errors in parentheses. *, **, and *** indicate significance at the 10, 5 and 1% levels respectively.
(a) denotes significant after applying the Benjamini-Hochberg procedure to maintain the false discovery rate across the four groups
at 10%. Square brackets show Lee upper and lower bounds, along with indicator of their significance.
All regressions include controls for course*gender*age group lotteries applied for.
Test of equality tests the null of equality of treatment impact across the four age*gender strata.
Appendix A defines the outcome variables.

Table 6: Heterogeneity with Respect to Pre-specified Individual Characteristics

		Differential Impact on:			
Treatment Interaction with:	Control Mean (Std. Dev.) of Interacting Variable	Employed 20+ hours	Aggregate Employment Index	Ever formal by Jan 12	Currently formal Aug 13
Expected benefit from ISKUR training	31.9	0.000	0.000	-0.000	0.000
	(27.3)	(0.001)	(0.001)	(0.000)	(0.000)

Post-high school education	0.434	0.014	0.013	0.009	-0.017
	(0.496)	(0.027)	(0.053)	(0.027)	(0.027)
Previously taken part in a training course	0.264	0.062**	0.099	0.030	0.033
	(0.441)	(0.031)	(0.061)	(0.031)	(0.031)
Has a child aged 6 or under	0.117	-0.047	-0.070	0.070*	-0.029
	(0.322)	(0.040)	(0.076)	(0.038)	(0.039)
Is the main decision-maker for whether they work	0.668	0.050*	0.104*	0.030	-0.041
	(0.471)	(0.028)	(0.053)	(0.028)	(0.028)
Raven test score	5.94	0.003	0.006	0.006	-0.004
	(3.10)	(0.004)	(0.009)	(0.004)	(0.004)
Numerate	0.564	0.003	0.034	0.013	-0.020
	(0.496)	(0.027)	(0.052)	(0.027)	(0.026)
Work centrality	4.08	0.023	0.052*	0.020	0.000
	(0.90)	(0.015)	(0.029)	(0.014)	(0.015)
Tenacity	8.37	0.010	0.018	0.010	-0.026***
	(1.30)	(0.011)	(0.020)	(0.010)	(0.010)
Unemployed for above the median duration	0.467	0.032	0.070	-0.004	0.015
	(0.499)	(0.027)	(0.053)	(0.027)	(0.027)

Notes: robust standard errors in parentheses, *, **, and *** indicate significance at the 10, 5 and 1% levels
Each row shows the treatment interaction from a regression which includes controls for course*age*gender lotteries, and for the level of the variable being interacted.

Table 7: Impact on Individual Well-Being and Household Well-Being

	Individual outcomes				**Household Outcomes**		
	Expected Prob. Of Working in 2 years	MHI-5 mental health (higher better)	Current Subjective Well Being	Subjective Well Being in 5 years	Household Income in last year	Transformed Household Income in last year	Durable Asset Index
ITT Estimate	0.923	-0.060	0.066*	0.061	485.524	0.057	0.106**
	(0.915)	(0.092)	(0.040)	(0.053)	(392.553)	(0.053)	(0.044)
Control Group Mean	54.1	18.508	4.436	5.822	21711	10.168	-0.071
Control Group Std. Dev.	33.4	3.410	1.514	1.980	14674	1.966	1.734
Sample Size	4878	5437	5508	5289	5396	5396	5495

Notes: Robust standard errors in parentheses. *, **, and *** indicate significance at the 10, 5 and 1% levels.

All regressions include controls for course*gender*age group lotteries applied for.
Appendix A defines the outcome variables.

Table 8: Comparison of Actual and Expected Treatment Impacts

	Employment Levels(%)		Treatment Impact (Proportion)		
	Expected percent chance of being employed if not trained (Std. dev)	Actual Control Employment Levels	ISKUR staff Expected Impact Mean (Std. dev)	Individual Expected Impact Mean (Std. dev)	Actual Impact LATE Estimate (std. error)
Full Sample	31.3	36.1	0.243	0.324	0.019
	(24.1)		(0.239)	(0.275)	(0.018)
Males under 25	35.4	40.7		0.297	-0.044
	(24.6)			(0.278)	(0.047)
Males 25 and over	33.5	54.8		0.302	0.081
	(26.1)			(0.280)	(0.054)
Females under 25	31.5	35.6		0.329	0.021
	(22.5)			(0.266)	(0.033)
Females 25 and over	27.7	24.7		0.346	0.022
	(23.5)			(0.277)	(0.027)

Notes: individual expectations are those of the treatment group at baseline.
Actual control employment level and LATE estimate for employment impact are for the outcome of working 20 hours or more per week.

Table 9: Which Course Characteristics are Associated with Better Impacts?

			Differential Impact on:			
Treatment Interaction with:	Sample Size	Control Mean (Std. Dev.) of Interacting Variable	Employed 20+ hours	Aggregate Employment Index	Ever formal by Jan 12	Currently formal Aug 13
Measures of course attendance rates						
Proportion assigned to course who attended it	5497	0.765	0.114	0.178	0.014	0.144

		(0.140)	(0.092)	(0.182)	(0.117)	(0.087)
Proportion assigned to course who completed it	5497	0.723	0.055	0.092	0.018	0.148*
		(0.164)	(0.083)	(0.168)	(0.095)	(0.081)
Proxies for course quality						
Course length above 320 hours	5494	0.423	-0.042*	-0.081*	-0.018	-0.024
		(0.494)	(0.024)	(0.048)	(0.027)	(0.023)
Average teacher experience greater than 12 months	4833	0.418	0.006	0.009	-0.017	0.030
		(0.493)	(0.026)	(0.052)	(0.029)	(0.025)
Percent of course teachers with tertiary education	4833	65.0	-0.000	-0.000	-0.000	-0.000
		(43.2)	(0.000)	(0.001)	(0.000)	(0.000)
Course has two or more competitors	4833	0.674	0.039	0.077	0.042	0.004
		(0.468)	(0.026)	(0.053)	(0.030)	(0.028)
Course offered by private provider	5494	0.348	0.044*	0.099**	0.062**	0.025
		(0.477)	(0.023)	(0.048)	(0.028)	(0.025)
Measures of what trainees thought course did						
Proportion who thought course taught new technical skills	5497	0.796	-0.087	-0.231	-0.134	-0.049
		(0.127)	(0.093)	(0.199)	(0.100)	(0.108)
Proportion who thought course certified skills they had already	5497	0.842	-0.012	-0.116	-0.062	0.093
		(0.103)	(0.108)	(0.225)	(0.104)	(0.107)
Proportion who thought course taught new job finding strategies	5497	0.604	-0.025	-0.101	0.000	0.073
		(0.147)	(0.090)	(0.182)	(0.108)	(0.084)
Proportion who thought course made them aware of new jobs	5497	0.449	-0.052	-0.166	-0.154**	0.007
		(0.183)	(0.072)	(0.148)	(0.068)	(0.060)
Measure of labour market demand						
Provincial unemployment rate is above median	5497	0.463	0.013	0.031	-0.001	0.049**
		(0.499)	(0.023)	(0.047)	(0.026)	(0.023)
Type of course						
Accounting course	5497	0.242	0.059**	0.136**	0.046	0.034
		(0.428)	(0.027)	(0.057)	(0.031)	(0.028)
Computer course	5497	0.153	0.002	0.019	0.080**	-0.039
		(0.360)	(0.033)	(0.067)	(0.038)	(0.035)

Notes: robust standard errors in parentheses, clustered at the course level. *, **, and *** indicate significance at the 10, 5 and 1% levels respectively.
Each row shows the treatment interaction from a regression which includes controls for course*age*gender lotteries, and for the level of the variable being interacted.

Appendix Table 1: What is Different about Private Courses?

	Public Provider	Private Provider
Course Characteristics		
Course length (days)	55	60***
Course length (hours)	328	353***
Accounting course	0.13	0.45***
Professional course	0.11	0.14***
Craftsman course	0.23	0.02***
Technical course	0.19	0.07***
Service course	0.24	0.14***
Course in Istanbul	0.27	0.12***
Average teacher experience greater than 12 months	0.45	0.34***
Percent of course teachers with tertiary education	71.9	51.2***
Individual Characteristics (administrative data)		
Female	0.61	0.66***
Age	27.0	26.5***
At least high school	0.66	0.85***
Individual characteristics (baseline data)		
Years of education	9.7	10.9***
Has done previous training course	0.25	0.20***
Household head	0.12	0.10**
Household size	3.86	3.64***
Married	0.30	0.27*
Ever employed	0.57	0.55
Total years working for pay	3.17	2.64***
Raven test score (out of 12)	6.10	5.93*
Numeracy score (out of 4)	3.31	3.51***
Tenacity	8.36	8.35

Note: Sample means shown.
*, **, and *** indicate statistically different from public providers at the 10, 5, and 1% levels respectively.